BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Selection: ELSE and Nested Conditions

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does selection do?

2. What does a structure diagram show?

A) The order instructions run
B) Every variable's value
C) The network's IP addresses
D) Parts and their hierarchy

3. "Doubling clock speed guarantees every program finishes in exactly half the time." Correct this. Explain why.

4. A user runs several apps. Some work can be shared across cores. "Each extra core doubles every program's speed." Correct this. When do more cores help?

Today's objectives

Use multiple branches and nested selection. Use else and elseif in a practical scenario.

Key vocabulary - Read, cover, recall

ELSE

An ELSE branch runs when its IF condition is false.

ELSEIF

An ELSEIF tests another condition after an earlier branch did not match.

Nested selection

An IF statement can be placed inside another to check a second condition.

Mutually exclusive branches

Only one matching route of an IF/ELSEIF/ELSE chain runs.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

What is ELSE?	ELSE - An ELSE branch runs when its IF condition is false.
In this example	A school attendance award uses elseif; An ELSEIF tests another condition after an earlier branch did not match.
Follow the effect	An IF statement can be placed inside another to check a second condition. Explain how this changes the eventual program result.
Check understanding	Mutually exclusive branches is used for a specific reason: Only one matching route of an IF/ELSEIF/ELSE chain runs.

I Do - One worked example

A school attendance award is being written. Explain how else and elseif are used to make the program work. [4 marks]

1. We are designing a school attendance award. Identify which data and decisions are essential.

2. ELSE [1] - An ELSE branch runs when its IF condition is false. [1].

3. ELSEIF [1] - An ELSEIF tests another condition after an earlier branch did not match. [1].

4. Two linked points for each technique give four mark points.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a online shop discount levels, explain how else helps the program. [2 marks]

2. For a theme park eligibility, explain why elseif matters. [2 marks]

3. When designing a competition medal bands, explain how nested selection and mutually exclusive branches should be used. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the comparison so the program reports insufficient storage.

```
freeGB = 4
neededGB = 6
if ... then
print("Enough space")
else
print("Not enough space")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A cafeteria age prices needs to work correctly. Explain how elseif and nested selection would help. [4 marks]

Q4. A pupil is programming a parking charge levels. Explain the roles of else, nested selection and mutually exclusive branches in their solution. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An ELSE branch runs when its IF condition is false. ☐ **T** ☐ **F**

2. ELSE - An ELSE branch runs when its IF condition is false. ☐ **T** ☐ **F**

3. A school attendance award uses elseif; An ELSEIF tests another condition after an earlier branch did not... ☐ **T** ☐ **F**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It chooses a branch using a condition. Selection can be nested. ☐

Q2 D - Parts and their hierarchy ☐

Q3 More clock cycles per second. This may let the CPU process more instructions if other factors are similar. ☐

Q4 More cores help when work can be split between them. Other CPU features and the task also affect the speed gained. ☐

Code First - Model

```
freeGB = 4
neededGB = 6
if freeGB >= neededGB then
print("Enough space")
else
print("Not enough space")
endif
```

Output: Not enough space

We Do - Check the model

1. ELSE [1] - An ELSE branch runs when its IF condition is false. [1]. ☐

2. ELSEIF [1] - An ELSEIF tests another condition after an earlier branch did not match. [1]. ☐

3. Nested selection [1] - An IF statement can be placed inside another to check a second condition. [1]. Mutually exclusive branches [1] - Only one matching route of an IF/ELSEIF/ELSE chain runs. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 ELSE [1]; Nested selection [1]. ☐

Q3 ELSEIF [1] - An ELSEIF tests another condition after an earlier branch did not match. [1]. Nested selection [1] - An IF statement can be placed inside another to check a second condition. [1]. ☐

Q4 ELSE [1] - An ELSE branch runs when its IF condition is false. [1]. Nested selection [1] - An IF statement can be placed inside another to check a second condition. [1]. Mutually exclusive branches [1] - Only one matching route of an IF/ELSEIF/ELSE chain runs. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: